Usability Study Report

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Application scope

Application description:

QuickPik is a lightweight matrix-based decision platform that helps groups quickly reach consensus. Users create a matrix poll with row and column options, participants vote on preferred combinations, and results are visualized as a heatmap for informed group decisions. The core problem QuickPik solves is the inefficiency of group decision-making through unstructured communication channels like group chats, where discussions spiral without resolution.

Users - Target audience:

Persona 1: The Indecisive Friend Group (“Emma”)

- Context: Emma and her friends are trying to decide where to eat dinner and when to go.
- Usage: Creates a matrix poll with five restaurant cuisines (Thai, Italian, Vietnamese, Mexican, Japanese) as rows and reservation times (6 PM, 7 PM, 8 PM) as columns.
- Pain point: Endless back-and-forth in group chats without reaching a clear decision.
- Outcome: The heatmap highlights the time-cuisine combination with the strongest group preference, helping Emma finalize the reservation quickly.

Persona 2: The Project Team Lead (“John”)

- Context: John is leading a student project team that needs to prioritize tasks for the next sprint.
- Usage: Creates a matrix poll with project tasks (UI Design, Database Setup, Authentication System, API Integration) as rows and priority levels (Low, Medium, High) as columns.
- Pain point: Manual prioritization through debates or informal Slack polls lacks structure.
- Outcome: The aggregated heatmap makes it clear which tasks the majority believes should be tackled first.

Persona 3: The Event Organizer (“Carla”)

- Context: Carla is planning a weekend outing for friends and needs to find the best activity and day.
- Usage: Creates a matrix poll with activity options (Hiking, Movie Night, Bowling, Dinner) and available days (Friday, Saturday, Sunday).
- Pain point: Coordinating availability and preferences across a group is time-consuming.
- Outcome: The heatmap reveals the activity-day combination that works best for most participants.

Data description:

QuickPik stores and manages data across the following MongoDB collections:

- Users — stores account information including username and password hash. Displayed in authentication and profile flows.
- Polls — stores poll metadata including title, row options, column options, creator reference, poll status (open/closed), and creation date. Displayed in the poll creation and management views.
- Votes — stores individual participant votes as cell selections within a poll matrix, along with the voter reference and timestamp. Displayed in the poll participation and heatmap visualization views.

Main tasks - use cases: (specific, measurable, concrete)

The following four tasks were used in the usability study:

Task	Description	Success Criteria
T1	App Exploration — Understand the purpose of the application by exploring the homepage freely.	User can articulate what QuickPik does after free exploration.
T2	Create a Poll — Create a matrix poll with a title, row options (activities), and column options (days), then find the sharable link.	Poll is created with all options visible in the matrix; sharable link/ID is located.
T3	Participate in a Poll — Join an existing poll via link/ID, vote on preferred matrix cells, change a vote, and remove all votes.	User can vote, toggle selections, and clear their submission.
T4	View Results & Manage Poll — View the heatmap visualization of aggregated votes, then close and delete the poll.	User can interpret the heatmap to identify popular combinations; poll is closed and deleted.

Experiment

Preparation

Introduction

- Welcome participants
- Ask for consent to record
- Ask to think out loud
- Remind them that they aren't being evaluated, and that they can leave at any time
- Ask participant to think like the target audience

Prepare demographics questions (offer an option to refuse to answer any of the questions. Only include relevant questions for the target audience)
Asked participant whether they are working professionally or students, in order to get them to think like either one of the user personas listed above (Project Team Lead/Indecisive Friend Group)

Prepare recording setup (audio, video and screen)
Utilized zoom meetings for the medium of the study to screen record user interactions and have an audio and video interaction

Prepare Script of the tasks to be read to participants:

- Script for intuitiveness: Given the link to the application, open the application and let me know what you think is the purpose of the application on a high level?
  ○Does the Help menu give a better clarity on the purpose of the application?●

- Script for Task T2: Create a poll
- ○Verify that all your decisions are visible on the poll matrix
  ○ Get the sharable link/poll-id for the poll and share it with me

- Script for Task T3: Participate in a poll
  ○ Verify that your decision is recorded (any visual confirmation)
  ○ What do the cells in the poll represent for this specific poll
  ○ Change your vote
  ○ Remove your vote

- Script for Task T4: Login back as the creator
  ○ What is the informed decision you will make after visualizing the results?
  ○ Close the poll

Post Usability Study Survey (Likert Scales)
● Task 1: How intuitive was it to understand the purpose of the application? (On a scale of 1-5)
  ○ 1 - Not at all intuitive (I could not understand what the app was for)
  ○ 5 - Extremely intuitive (The purpose was immediately clear)

● Task 2: How effective was the application in capturing the various choices for the decision that you wanted to make and sharing the poll with others?
  ○1 - Not at all effective (choices were clear by just looking at the matrix poll and sharing the poll was difficult)
  ○ 5 - Extremely effective (choices looked confusing when displayed in the matrix poll and sharing the poll was easy)

● Task 3: How intuitive was the application for you to be able to participate in a poll (including editing/removing your response)?
  ○ 1 - Not at all intuitive (could not understand the choices in the matrix format and was not able to edit/delete responses)
  ○5 - Extremely intuitive (could easily understand the choices and participate in the

poll and was also able to edit/remove my response)

- Task 4: How effective was the application in visualizing the results of the poll?
  - 1 - Not at all effective (could not understand at all what choices got the majority votes)
  - 5 - Extremely effective (was easily able to understand the majority and minority choices and take an informed decision based on the result)

- Any other feedback for the application

Experiment Notes

*After finishing running the experiment for each participant, make sure to write down right away all the notes you remember from the session. Then, **rewatch the session** video looking for parts on which the participants were frustrated or struggling to complete a task, and write notes for that. Using that information identifies issues, bugs and other points of improvements.*

Participant 1

Demographics

- Student pursuing a Master's degree
- Followed the **"Event Organizer"** persona

Detailed Notes

Task T1 – Intuitiveness

- User did not intuitively understand the purpose of the application from the landing page.
- Navigated to the Help menu but got distracted due to overly detailed steps.
- Returned to the landing page and explored buttons and fields to infer functionality.

Observation:

- The user struggled to understand the purpose of the application even after checking the Help section.
- Required intervention to interpret UI fields and their meaning.

Task T2

- User initially struggled to understand what the **"Rows"** and **"Columns"** fields represented.

Observation:

- After initial confusion, the user quickly adapted.
- Successfully created, verified, and shared the poll link.

Task T3

- User pasted the poll link into the browser URL.
- Participated by selecting appropriate cells.
- Modified responses by unchecking some cells and updating the selection.

Observation:

- User was able to participate in the poll smoothly and without difficulty.

Task T4

- User logged back in using poll creator credentials.
- Analyzed results using the heatmap.
- Closed the poll after reviewing outcomes.

Observation:

- User effectively visualized results and made an informed decision to close the poll.

Post-Test Questionnaire Results (Likert Scale)

- **T1 Score:** 4
- **T2 Score:** 5
- **T3 Score:** 5
- **T4 Score:** 4

VIDEO LINK - https://www.youtube.com/watch?v=6xRbWv7X-io&feature=youtu.be

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Participant 2

Demographics

- Student pursuing a Master's degree
- Followed the "Project Team Lead" persona

Detailed Notes

Task T1 – Intuitiveness

- User explored the landing page and quickly identified the "Create a Poll" and "Join a Poll" buttons.
- Briefly hovered over the Help section but did not open it.
- Correctly inferred the app was for group decision-making based on the homepage text and layout.

Observation:

- The user was able to understand the general purpose of the application from the landing page without intervention, though noted the description could be more prominent.

Task T2

- User clicked the "Create a Poll" button and navigated to the form without difficulty.
- Filled in the title, row options, and column options with minimal hesitation.
- Located the sharable poll ID but expressed that copying it felt manual and wished for a copy button.

Observation:

- Poll creation was smooth. The main friction point was sharing — the user expected a one-click copy mechanism for the poll ID.

Task T3

- User entered the poll ID in the join field to access the researcher's poll.
- Voted on preferred cells and noticed the visual feedback (color change) confirming selections.
- Successfully toggled a vote off and selected a different cell.
- Attempted to remove all votes but took a moment to locate the remove/clear option.

Observation:

- Voting and editing were intuitive. Removing the entire submission required some exploration but was completed without intervention.

Task T4

- User logged back in with the creator credentials and navigated to the poll.
- Interpreted the heatmap correctly, identifying the most popular combination at a glance.
- Closed the poll and then deleted it using the options menu.

Observation:

- The heatmap was immediately clear to the user. Closing and deleting the poll were completed without confusion.

Post-Test Questionnaire Results (Likert Scale)

- T1 Score: 5
- T2 Score: 5
- T3 Score: 4
- T4 Score: 4

Open-ended feedback: "Poll ID in a format to copy to the clipboard. Along with the link, a poll ID. Then more about the poll names on row and column."

VIDEO LINK - https://www.youtube.com/watch?v=ZJy4GoG4sAI

Participant 3

Demographics

- Student pursuing a Master's degree
- Followed the "Indecisive Friend Group" persona

Detailed Notes

Task T1 – Intuitiveness

- User spent time reading the landing page content carefully before interacting with any elements.
- Clicked on the Help section and read through the step-by-step instructions.
- After reviewing the help content, returned to the main page and expressed a clear understanding of the app's purpose.

Observation:

- The user relied on the Help section to form an understanding. While they ultimately grasped the purpose, the landing page alone was not sufficient — the Help content filled the gap.

Task T2

- User navigated to the poll creation form and filled in all fields.
- Initially typed placeholder-style text before correcting to actual activity and day names, suggesting the generic "Row 1" / "Column 1" placeholders were not guiding input well.
- Successfully created the poll and found the sharable link.

Observation:

- The creation flow was functional but the placeholder text caused a brief moment of hesitation. Contextual examples would have helped the user fill in fields faster.

Task T3

- User pasted the shared link into the browser URL bar to join the poll.
- Voted on multiple cells and understood the toggle behavior on first try.

- Changed a vote by deselecting one cell and selecting another.
- Removed all votes by deselecting each cell individually — did not find a bulk clear option.

Observation:

- Participation was smooth overall. The lack of a "Clear All" button meant the user had to manually deselect each cell, which was slightly tedious.

Task T4

- User logged back in and navigated to the created poll.
- Reviewed the heatmap and correctly identified the winning combination.
- Closed the poll but noted there was no visible status change indicating the poll was now closed.
- Deleted the poll after locating the option in the menu.

Observation:

- The heatmap was effective. The user flagged that after closing the poll, there was no visual indicator confirming the closed state — this could confuse creators checking back later.

Post-Test Questionnaire Results (Likert Scale)

- T1 Score: 3
- T2 Score: 4
- T3 Score: 4
- T4 Score: 4

Open-ended feedback: "Add a copy button to copy the poll ID easily. Add a sample of what the poll looks like."

VIDEO LINK - https://www.youtube.com/watch?v=DCT-N3ynN-0

Prioritized list of issues and corresponding changes:

Summarize the most crucial issues and changes proposed

Issue 1: No copy-to-clipboard button for Poll ID
Change: Add a one-click “Copy to Clipboard” button next to the poll ID/link. Provide visual feedback (e.g., “Copied!” tooltip) on successful copy.
Priority: Must
Was it implemented? How? Pending - identified as a priority fix.

Issue: Matrix concept not immediately clear to first-time users
Change: Add a brief instructional tooltip, onboarding modal, or inline example on the poll page explaining that each cell represents a combination of a row option and column option, and that clicking a cell registers a vote.
Priority: Should
Was it implemented? How? Pending - identified as a priority fix.

Issue: Overall visual appeal needs polish
Change: Refine typography, spacing, color palette, and component styling for a more polished look. Consider consistent use of Bootstrap components and a cohesive design system.
Priority: Could
Was it implemented? How? Pending - identified as a possible fix.

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Usability Study Report

Author: Alex Joseph

Link to the studies:
Participant #1: https://www.youtube.com/watch?v=WDDjyAOSEk4
Participant #2: https://www.youtube.com/watch?v=wx4p7cg92-I
Participant #3: https://www.youtube.com/watch?v=b6nVG5Vfk50

Application scope

Application description: In real-world group discussions regarding decisions like - picking a restaurant, scheduling a meeting, prioritising tasks - stall because everyone has slightly different preferences and there is no clear way to see where the group agrees. Creating a simple poll is often not sufficient, as it typically forces users to choose a single option and fails to capture nuanced preferences across multiple dimensions, leaving groups without a clear, consensus-driven outcome.

QuickPik solves this by letting a creator build a decision matrix poll: rows and columns represent the options, participants check off their preferred combinations, and the system aggregates all responses into a visual heatmap. The heatmap instantly reveals the cell(s) with the strongest collective agreement, turning a messy group chat into a clear, data-driven decision.

Users - Target audience (User Personas):
1. The Indecisive Friend Group ("Karthik"):
Emma and her friends are trying to decide where to eat dinner and when to go. Instead of debating endlessly in a group chat, Emma creates a QuickPik matrix poll with five restaurant cuisines (Thai, Italian, Vietnamese, Mexican, Japanese) as rows and possible reservation times (6 PM, 7 PM, 8 PM) as columns. Each friend marks the combinations they are comfortable with. The resulting heatmap visualization highlights the time and cuisine combination with the strongest group preference, helping Emma quickly finalize the reservation.

2. The Project Team Lead ("Sesmi"):
Daniel is leading a student project and the team needs to decide which tasks should be prioritized for the next sprint. He creates a QuickPik matrix poll listing project tasks as rows (UI Design, Database Setup, Authentication System, API Integration) and priority levels as columns (Low, Medium, High). Each team member marks how important they believe each task is. The aggregated results produce a *priority heatmap*, making it immediately clear which tasks the majority of the team believes should be tackled first.

3. The Event Organizer ("Carla"):
Sofia is planning a weekend outing for her friends and needs to determine the best activity and day. She creates a QuickPik matrix poll with activity options (Hiking, Movie Night, Bowling, Dinner) and available days (Friday, Saturday, Sunday). As friends submit their availability and

preferences, the heatmap clearly reveals the activity-day combination that works best for most participants.

Data description: (what data is stored and displayed)
Data for the application is stored across 4 entities:

1. Users: stores in each record, basic user information like firstname, lastname, email and password.
2. Polls: stores in each record, the metadata associated with a created poll like title, description, created date, column and row headers.
3. Submissions: stores in each record, the poll the submission is associated with and the selected cells of the poll matrix for that poll.
4. Session: stores the session history for the application.

Data displayed:

1. A logged in user can see all the polls that they have created and on clicking a poll they can see all the submissions for that poll in the form of a heatmap.
2. A logged in user can go to the specific poll id and see their submission (marked cells) in the participant page.

Main tasks - use cases:

- T1: Understand the purpose of the app (Inspect landing page, get clarity via help menu)
- T2: Create a poll, verify the matrix decisions and share the link
- T3: Participate in the created poll, interpret cells, modify and remove vote
- T4: Review results as creator, make the decision and close the poll

Experiment

Preparation

Introduction

- Welcome participants
- Ask for consent to record
- Ask to think out loud
- Remind them that they aren't being evaluated, and that they can leave at any time
- Ask participant to think like the target audience

Prepare demographics questions (offer an option to refuse to answer any of the questions. Only include relevant questions for the target audience)

Asked participant whether they are working professionally or students, in order to get them to think like either one of the user personas listed above (Project Team Lead/Indecisive Friend Group)

Prepare recording setup (audio, video and screen)

Utilized zoom meetings for the medium of the study to screen record user interactions and have an audio and video interaction

Prepare Script of the tasks to be read to participants:

- Script for intuitiveness Task T1: Given the link to the application, open the application and let me know what you think is the purpose of the application on a high level?
    - Does the Help menu give a better clarity on the purpose of the application?
- Script for Task T2: Create a poll
    - Verify that all your decisions are visible on the poll matrix
    - Get the sharable link/poll-id for the poll and share it with me
- Script for Task T3: Participate in a poll
    - Verify that your decision is recorded (any visual confirmation)
    - What do the cells in the poll represent for this specific poll
    - Change your vote
    - Remove your vote
- Script for Task T4: Login back as the creator
    - What is the informed decision you will make after visualizing the results?
    - Close the poll

Post Usability Study Survey (Likert Scales)

- Task 1: How intuitive was it to understand the purpose of the application? (On a scale of 1-5)
    - 1 - Not at all intuitive (I could not understand what the app was for)
    - 5 - Extremely intuitive (The purpose was immediately clear)
- Task 2: How effective was the application in capturing the various choices for the decision that you wanted to make and sharing the poll with others?
    - 1 - Not at all effective (choices were clear by just looking at the matrix poll and sharing the poll was difficult)
    - 5 - Extremely effective (choices looked confusing when displayed in the matrix poll and sharing the poll was easy)
- Task 3: How intuitive was the application for you to be able to participate in a poll (including editing/removing your response)?
    - 1 - Not at all intuitive (could not understand the choices in the matrix format and was not able to edit/delete responses)
    - 5 - Extremely intuitive (could easily understand the choices and participate in the poll and was also able to edit/remove my response)
- Task 4: How effective was the application in visualizing the results of the poll?
    - 1 - Not at all effective (could not understand at all what choices got the majority votes)
    - 5 - Extremely effective (was easily able to understand the majority and minority choices and take an informed decision based on the result)
- Any other feedback for the application

Experiment Notes

After finishing running the experiment for each participant, make sure to write down right away all the notes you remember from the session. Then, ***rewatch the session*** *video looking for*

parts on which the participants were frustrated or struggling to complete a task, and write notes for that. Using that information identifies issues, bugs and other points of improvements.

Participant 1 (https://www.youtube.com/watch?v=WDDjyAOSEk4)
Demographics answers: Student pursuing Master's (field was not mentioned) - followed the "Event Organizer" persona.
Detailed notes

- Notes for intuitiveness Task T1: User did not intuitively understand the purpose of the application through the landing page; proceeds to help menu, but gets distracted because of the detailed steps; navigates back to landing page to understand the purpose of the app by checking out the buttons and fields.
    - The user failed in understanding the purpose of the application, despite checking out the help me and required intervention to understand what the fields in the UI represented.
- Notes for Task T2: User struggled to understand what the "rows" and "columns" fields represented.
    - Although the user initially struggled a bit to understand what the "rows" and "columns" fields represented, user quickly caught up and was able to create, verify and share the poll link successfully.
- Notes for Task T3: User pasted the link in the browser url; participated in the poll by selecting the appropriate cells; modified the response by unchecking a few cells and updating response.
    - User was able to participate in poll successfully with ease.
- Notes for Task T4: User logged back in with poll creator credentials; analyzed the results through the heatmap; closed the poll.
    - User was able to visualize the results, make the informed decision and close the poll.

Post-test questionnaire results (Likert scales results attached below):

- Score for T1: 4
- Score for T2: 5
- Score for T3: 5
- Score for T4: 4

Participant 2 (https://www.youtube.com/watch?v=wx4p7cg92-I)
Demographics answers: Student pursuing Master's in Computer Science, has a friends group - followed "The Indecisive Friend Group" persona.
Detailed notes

- Notes for intuitiveness Task T1: User was able to sign up and log in without any issues; Upon accessing the application, read the placeholder values and observing the UI elements, the user inferred that the matrix style decision making application.
    - user demonstrated a clear understanding of the purpose of the application;
- Notes for T2: User understood how to utilize the "rows" and "columns" fields while creating the poll.
    - User interpreted the fields correctly and created a poll using meaningful categories (e.g., locations vs. seasons).

- Notes for T3: User attempted to paste the poll link into the “Join Poll” input box; required minor intervention to understand that the link should instead be pasted into the browser URL; user checked the cells to participate; modified selection to update response.
    - After the clarification on how to reach the participant page of the poll, the user was able to proceed.
- Notes for T4: User logged back in as the poll creator; analyzed the poll results using the heatmap visualization; interpreted results; closed poll
    - User was able to interpret the leading votes effectively and proceeded to close the poll without any issues.

Post-test questionnaire results (Likert scales):

- Score for T1: 3
- Score for T2: 4
- Score for T3: 2
- Score for T4: 3.

Participant 3 (https://www.youtube.com/watch?v=b6nVG5Vfk50)

Demographics answers: Student pursuing Master’s (field not mentioned), has a friends group - followed “The Indecisive Friend Group” persona.

Detailed notes

- Notes for intuitiveness Task T1: User initially misinterpreted the “rows” and “columns” fields as headers for a list-based or tabular data entry application.
    - User struggled to understand the actual purpose of the application as the labels for the “rows” and “columns” fields were not intuitive.
- Notes for T2: User could not figure out what the “columns” represented while creating the poll.
    - User required guidance in understanding that the columns add nuance to the “rows” of the poll, and then they were able to create a successful poll for deciding “weekend movie plans”.
- Notes for T3: User attempted to paste the poll link into the “Join Poll” input box; later figured to extract poll id from the url; pasted poll id into the “join poll” box.
    - After reaching the participant page of the poll, the user was able to proceed by selecting their preferred movie choices on their preferred days. User also successfully modified their response.
- Notes for T4: User logged back in as the poll creator; analyzed the poll results using the heatmap visualization; interpreted results; struggled with the close poll button.
    - User was able to interpret the leading votes effectively but had difficulty in understanding the purpose of the “close poll” button.

Post-test questionnaire results (Likert scales):

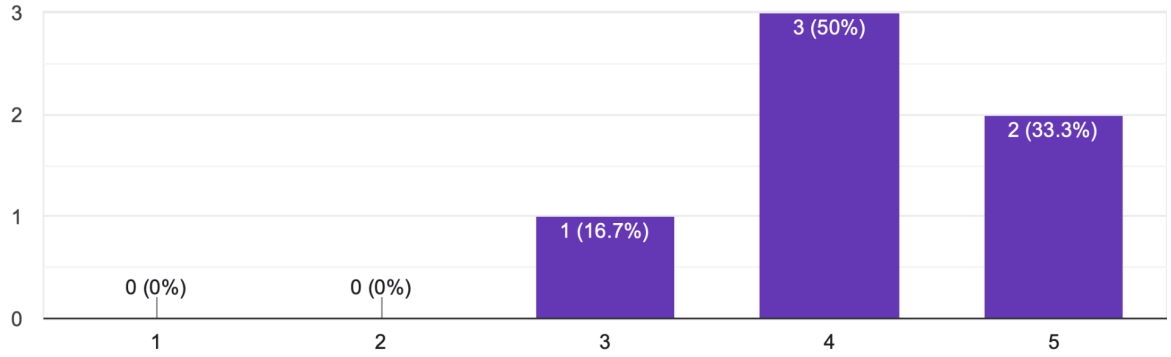
- Score for T1: 4
- Score for T2: 5
- Score for T3: 5
- Score for T4: 5

Post Usability Study Survey Results

Please note: the likert scale survey questions are accumulated over all 6 participants

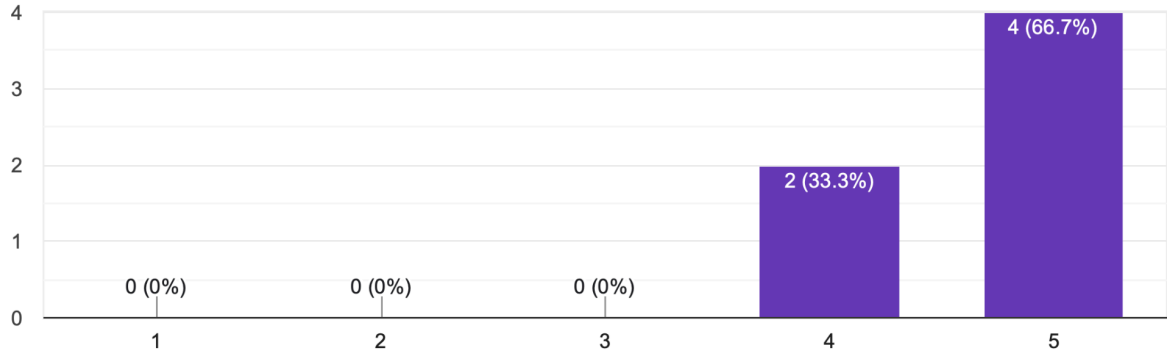
Task 1: How intuitive was it to understand the purpose of the application? (On a scale of 1-5)

6 responses



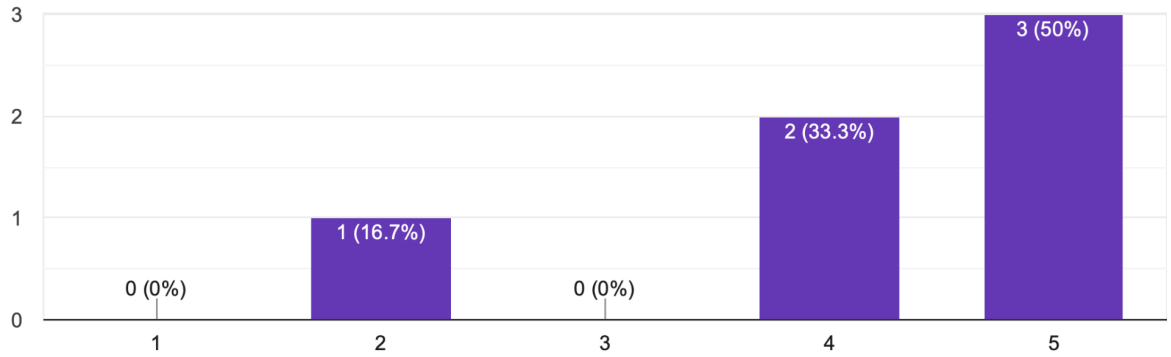
Task 2: How effective was the application in capturing the various choices for the decision that you wanted to make and sharing the poll with others?

6 responses



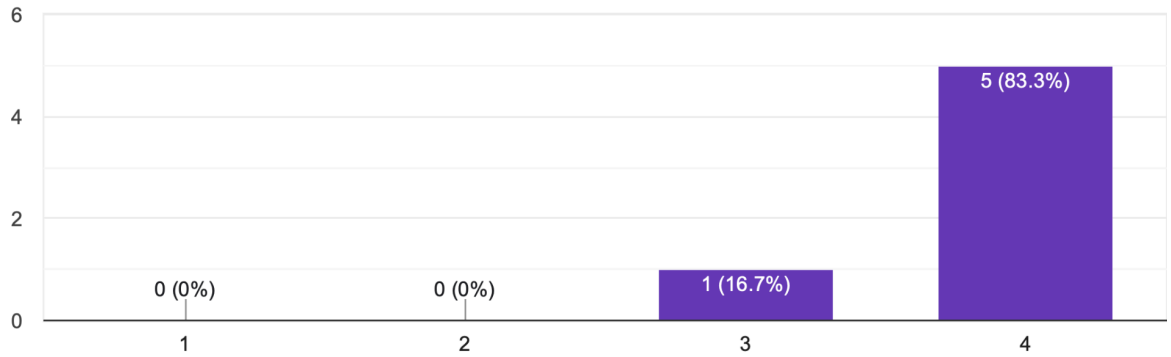
Task 3: How intuitive was the application for you to be able to participate in a poll (including editing/removing your response)?

6 responses



Task 4: How effective was the application in visualizing the results of the poll?

6 responses



Prioritized list of issues and corresponding changes:

Summarizing critical issues pointed by users:

Issue: Difficulty in understanding what the “Rows” and “Columns” represent.
Change: As per feedback suggested by the users - there could be an example for a pre-filled poll or a gif animation filling up sample values for the rows and columns, or more intuitive labels instead of “Rows” and “Columns”
Priority: Must have priority, as this was a common issue faced by all 3 participants

Issue: Struggle in reaching the participant page for the specific poll created.
Change: As per the feedback suggested by the users - it should be clear as to what to paste into the “Join Poll” input field.

Priority: Should have priority as 2 out of the 3 users faced difficulty in trying to reach the participant page.

Issue: Remove user response option.
Change: One user wanted to remove his response in the poll, and the only option was to revert the cells and updated the response. Rather there should be a more explicit “remove response” option.
Priority: Should have priority as the reverting selected cells for a large poll can become extremely difficult for the users.